How Much Biological Detail Does a Neuron Model Need?

Evaluating Conductance-Based Multi-Compartment versus Dimensional FitzHugh–Nagumo Reductions Across Four Mouse Visual Cortex Populations

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Abstract

A central question in computational neuroscience is how much biological detail a neuron model requires to reproduce electrophysiological behaviour, and which computational features are preserved or lost when high-dimensional conductance-based models are reduced to low-dimensional phenomenological models. We investigated this question by conducting a systematic, three-way empirical comparison across four layer-defined spiny excitatory neuronal representatives from mouse primary visual cortex (VISp layers 4, 2/3, 5, and 6a) from the Allen Cell Types Database. We compared somatic intracellular current-clamp recordings against: (1) published, frozen perisomatic multi-compartment biophysical models simulated in NEURON containing passive leak and up to seven active conductances (Na_v, K_v, K_v3, K_{slow}, I_A, I_h, and Ca); and (2) two-variable dimensional FitzHugh–Nagumo (FHN) models parameterized by eight biophysical quantities ($C_m, g_L, V_{\mathrm{rest}}, V_{\mathrm{th}}, V_{\mathrm{peak}}, k_v, \tau_w, a_w$) calibrated on a single current sweep per population.

Models were evaluated out-of-sample on 109 previously held-out experimental sweeps (97 unique current levels). Across the held-out dataset, biological recordings exhibited action potentials in 40 sweeps, the detailed models fired in 24 sweeps, and the FHN models fired in 44 sweeps (37 True Positives, 7 False Positives, 3 False Negatives, 62 True Negatives; sensitivity 92.5%, specificity 89.9%). FHN matched passive subthreshold charging with high fidelity ($r^2 > 0.996$ on hyperpolarizing sweeps) and accurately detected near-threshold excitability transitions through an analytically verified subcritical Andronov–Hopf bifurcation with numerically confirmed open bistability intervals across all four populations: A (214.16, 234.34) pA, B (146.72, 161.28) pA, C (360.03, 385.33) pA, and D (215.92, 236.27) pA.

However, model reduction to two planar variables resulted in four distinct limitations: (1) an inability to sustain multi-second spike-frequency adaptation (observed across all 23 qualifying biological multi-cycle sweeps, with median inter-spike interval expansions of $1.97\times$ –$3.45\times$), as autonomous FHN limit cycles contract to stationary periodic firing within ≤ 2 cycles; (2) substantial step-waveform error (step RMSE 12.36–23.73 mV vs. 4.15–8.30 mV in detailed models) driven by broad cubic relaxation oscillations compared to fast biological spikes; (3) an empirical trade-off between delayed onset latency and sustained repetitive firing across 961 tested parameter combinations in Layer 6a, where parameter sets achieving biological repetitive rates (≥ 12 spikes) collapsed latency to ≤ 17.40 ms; and (4) absence of hyperpolarization-activated inward sag ($\Delta V_{\mathrm{sag}} = 0.00$ mV in FHN vs. 0.63–1.86 mV biological). Conversely, detailed models exhibited vulnerability to parameter-tuning artifacts: an uncalibrated +13.6 mV resting offset in Layer 4 accounted for 92.3% of its overall mean squared error, and calibration on suprathreshold sweeps correlated with a +60 pA rheobase overestimation in Layers 4 and 5.

We conclude that the tested two-variable dimensional FHN formulation is sufficient for passive subthreshold integration and macroscopic threshold gating, but cannot reproduce spike-train timing, multi-second rate adaptation, or waveform kinetics under the evaluated parameterizations, providing strong scientific motivation for exploring modelling extensions such as secondary recovery variables or calcium-dependent adaptation processes.

1. Introduction

Mathematical models of single neurons span a broad spectrum of abstraction, from highly detailed multi-compartment models that incorporate complex dendritic morphologies and tens of spatially distributed voltage-gated ion channels, to low-dimensional phenomenological models that capture the geometry of excitability with two coupled ordinary differential equations (ODEs). Choosing the appropriate level of abstraction is a foundational problem in computational neuroscience. Highly detailed models offer biophysical interpretability and allow direct pharmacological manipulation of specific conductances, but they require high computational overhead and suffer from parameter indeterminacy and calibration complexity. Conversely, low-dimensional models, such as the FitzHugh–Nagumo (FHN) model or the Integrate-and-Fire family, provide analytical tractability, geometric phase-plane intuition, and computational scalability suitable for large-scale network simulations, but necessarily compress multiple biophysical mechanisms into a few phenomenological variables.

The central scientific challenge is determining what neurocomputational capabilities are preserved and what behaviours are irrecoverably lost when a biological neuron is abstracted into a low-dimensional model. While theoretical reductions—such as asymptotic reductions of the Hodgkin–Huxley equations to planar systems via timescale separation and fast-slow analysis—are mathematically well-established, quantitative out-of-sample evaluations of such models against real intracellular mammalian recordings across diverse cortical layers remain scarce. Furthermore, comparative studies frequently compare models on different datasets or evaluate models on the same data used for calibration, conflating parameter-fitting flexibility with predictive validity.

In this study, conducted as part of the Introduction to Neural and Cognitive Modelling (INCM) research program, we present an empirical evaluation of model reduction. We systematically compare three tiers of evidence: (1) intracellular somatic recordings from four layer-defined spiny excitatory neuronal representatives from the primary visual cortex (VISp) of adult mice; (2) frozen, published perisomatic multi-compartment biophysical models developed by the Allen Institute for Brain Science; and (3) dimensional two-variable FitzHugh–Nagumo models calibrated on a single stimulus sweep per population. By isolating 109 held-out experimental sweeps encompassing diverse stimulus protocols, amplitudes, and durations, we provide an objective, cross-population benchmark of the limits and capabilities of two-variable neuron models.

2. Research Question and Specific Objectives

2.1 The Central Research Question

The core scientific question guiding this research is:

> ***"How much biological detail does a neuron model need to reproduce a real neuron's electrophysiological behaviour, and which behaviours are preserved or lost when a detailed conductance-based model is reduced to a FitzHugh–Nagumo model?"***

2.2 Specific Research Objectives

To address this question empirically and mathematically, we pursued four specific objectives:

1. **Pipeline and Data Readiness:** Validate and curate somatic current-clamp electrophysiology recordings from the Allen Cell Types Database across four layer-defined VISp spiny excitatory populations (Layer 4, Layer 2/3, Layer 5, and Layer 6a), ensuring rigorous Neurodata Without Borders (NWB) format compliance and physical unit consistency.
2. **Biophysical Baseline Execution:** Execute and benchmark the frozen, published Allen perisomatic multi-compartment models in the NEURON simulation environment on identical stimulus sweeps, establishing the detailed conductance-based baseline.
3. **Dimensional FHN Calibration:** Formulate a physically dimensional two-variable FitzHugh–Nagumo model parameterized by measurable biophysical quantities ($C_m, g_L, V_{\text{rest}}, V_{\text{th}}, V_{\text{peak}}, k_v, \tau_w, a_w$), and calibrate it on a single representative current-clamp sweep per population using a dual-objective optimization algorithm.
4. **Held-Out Evaluation and Dynamical Audit:** Evaluate both model tiers out-of-sample across 109 held-out sweeps, perform independent mathematical bifurcation analysis (computing first Lyapunov coefficients and limit-cycle fold boundaries), and quantify behaviours across a five-tier taxonomy of model reduction fidelity.

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3. Biological and Mathematical Background

3.1 Electrophysiological Properties of Cortical Pyramidal Neurons

Neocortical spiny excitatory neurons exhibit complex, non-linear membrane dynamics in response to somatic current injection:

- **Passive Subthreshold Charging:** In the hyperpolarizing and weak depolarizing regimes, current flows primarily through passive potassium and chloride leak channels, yielding an exponential charging curve governed by the membrane time constant $\tau_m = R_{\text{in}} C_m$.
- **Inward Rectification (Sag):** Strong hyperpolarization activates hyperpolarization-activated cyclic nucleotide-gated (HCN) non-specific cation channels, generating an inward current (I_h) that depolarizes the membrane toward rest during prolonged pulses, producing a characteristic "sag" (ΔV_{sag}).
- **Threshold Excitability:** Depolarization beyond a threshold voltage triggers rapid activation of voltage-gated sodium channels (Na_v), generating action potentials followed by delayed-rectifier potassium channel (K_v) activation that repolarizes the membrane.
- **Spike-Frequency Adaptation (SFA):** During sustained depolarizing steps, cortical pyramidal cells typically exhibit progressive lengthening of successive inter-spike intervals (ISIs), mediated by slow calcium-activated potassium currents (I_{AHP}, I_{SK}) and non-inactivating muscarinic-sensitive potassium currents (I_M).
- **Delayed Firing:** Specific cortical populations, particularly in deep layers (Layer 6a), often display delayed first-spike onset latencies ($t_{\text{lat}} > 50$ ms) near threshold, classically associated in the literature with slowly inactivating transient A-type potassium currents (I_A).

3.2 The Hodgkin–Huxley and Perisomatic Conductance Framework

The biophysical conductance-based framework, pioneered by Hodgkin and Huxley (1952), describes membrane potential dynamics by summing individual ionic currents across the lipid bilayer:

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -\sum_i \bar{g}_i m_i^p h_i^q (V - E_i) - g_L (V - E_L) + I_{\text{inj}}(t)$$

where $\bar{g}_i$ represents maximal channel conductance, E_i is the Nernst equilibrium potential for ion species i, and m_i, h_i are activation and inactivation gating variables governed by voltage-dependent first-order kinetics:

$$\frac{\mathrm{d}m_i}{\mathrm{d}t} = \frac{m_{\infty,i}(V) - m_i}{\tau_{m,i}(V)}$$

In multi-compartment perisomatic models (Gouwens et al., 2018), the cable equation is discretized across reconstructed dendritic and axonal geometries, with active conductances placed at the soma and axon initial segment. This framework requires integrating dozens of coupled non-linear ODEs alongside spatial cable equations.

3.3 The FitzHugh–Nagumo Reduction

FitzHugh (1961) and Nagumo et al. (1962) demonstrated that the four-variable Hodgkin–Huxley model can be reduced to a planar dynamical system by exploiting timescale separation. Because sodium activation (m) operates on a sub-millisecond timescale ($\tau_m < 0.1$ ms), m can be replaced by its steady-state function $m_\infty(V)$. Furthermore, sodium inactivation (h) and potassium activation (n) evolve on similar, slower timescales and exhibit complementary voltage dependencies ($h \approx 1 - n$). Grouping these into a single effective recovery variable w yields a two-variable planar vector field:

$$\frac{\mathrm{d}V}{\mathrm{d}t} = F(V) - w + I_{\text{inj}}$$
$$\frac{\mathrm{d}w}{\mathrm{d}t} = \epsilon (G(V) - w)$$

where $F(V)$ is a non-monotonic cubic nullcline capturing regenerative upstroke and downstroke dynamics, and $G(V)$ is a monotonic linear recovery nullcline.

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4. Dataset and Specimen Selection

4.1 The Allen Cell Types Database

Electrophysiological data were curated from the publicly available Allen Cell Types Database (Allen Institute for Brain Science, 2015). Recordings were obtained via whole-cell patch-clamp electrophysiology from visually identified neurons in acute brain slices of adult mouse primary visual cortex (VISp), recorded at a sampling rate of 200 kHz ($5\,\mu$s sampling interval) at physiological temperature ($34°$C).

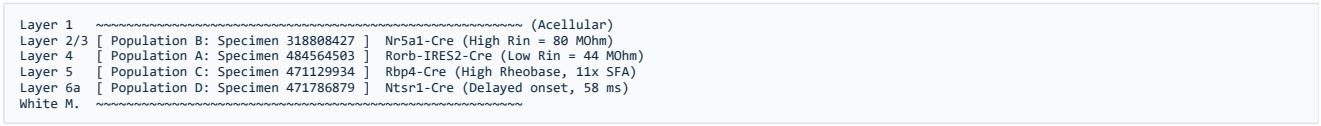
4.2 Four Layer-Defined VISp Populations

Rather than pooling heterogeneous cells across cortical layers, we selected four representative spiny excitatory neurons, each corresponding to an anatomically defined laminar group in VISp:

Table 1: Specimen Profiles and Baseline Electrophysiology Across Four Layer-Defined Populations

Population	Cortical Layer	Specimen ID	Transgenic Cre Line	Dendritic Morphology	Input Resistance (R_{in})	Resting Potential (V_{rest})	Primary Calibration Sweep
A	**Layer 4**	`484564503`	Rorb-IRES2-Cre	Spiny (Pyramidal)	44.3 MΩ	-73.73 mV	Sweep 49 (+370 pA, COARSE)
B	**Layer 2/3**	`318808427`	Nr5a1-Cre	Spiny (Pyramidal)	80.3 MΩ	-78.96 mV	Sweep 40 (+190 pA, COARSE)
C	**Layer 5**	`471129934`	Rbp4-Cre_KL100	Spiny (Pyramidal)	31.3 MΩ	-77.34 mV	Sweep 51 (+410 pA, COARSE)
D	**Layer 6a**	`471786879`	Ntsr1-Cre_GN220	Spiny (Pyramidal)	52.6 MΩ	-75.71 mV	Sweep 42 (+250 pA, COARSE)

Note: Data curated from the Allen Cell Types Database. All specimens are spiny excitatory neurons recorded in adult mouse VISp. Primary calibration sweeps represent coarse current injections used for single-sweep parameter fitting.



4.3 Crucial Scope Clarification: Layer-Defined Groups vs. Projection Classes

A vital methodological boundary must be emphasized: **these four groups are defined strictly by anatomical cortical layer and spiny morphology, not by confirmed axonal projection classes.** In cortical physiology, pyramidal neurons in Layers 5 and 6 exhibit distinct functional phenotypes depending on projection targets (e.g., Intratelencephalic [IT], Pyramidal Tract [PT], Extratelencephalic [ET], or Corticothalamic [CT]). The Allen Cell Types metadata does not contain retrograde tracer or axonal projection tracking for these specimens. Therefore, our findings characterize layer-specific representative specimens rather than broad, homogenous projection classes. Furthermore, because each population is represented by a single biological specimen ($n = 1$), our results reflect detailed case studies of layer-specific electrophysiology rather than statistical population distributions across all neurons in a cortical layer.

4.4 Stimulation Protocols and Data Isolation

Recordings evaluated in this project belong to the **Long Square** current-clamp protocol family:

- **Step Duration:** 1000 ms constant-amplitude current step.
- **Total Sweep Duration:** Variable, typically 1140–8000 ms, including pre-stimulus baseline (1020 ms) and post-stimulus recovery.
- **Protocol Sub-types:**
- `COARSE` (`Long Square`): Coarse current steps with amplitude increments of $\Delta I = 20$ pA, spanning deeply hyperpolarizing (-110 pA) to strongly suprathreshold ($+430$ pA) regimes.
- `FINEST` (`Long Square - Subthreshold / Fine`): Fine current steps with amplitude increments of $\Delta I \leq 10$ pA focused tightly around rheobase.
- **Held-Out Isolation:** For each population, exactly **one** suprathreshold COARSE sweep was designated as the primary calibration sweep. All remaining sweeps—amounting to **109 sweeps across the four specimens** (31 in Pop A, 18 in Pop B, 34 in Pop C, and 26 in Pop D)—were placed in strict quarantine and evaluated only after models were frozen bit-for-bit.

5. Detailed Allen Biophysical Models

5.1 Architecture and Conductance Manifest

The detailed models evaluated in this study are the authoritative, frozen perisomatic active models generated by the Allen Institute (Gouwens et al., 2018). These models were simulated in the NEURON simulation environment (version 8.2.7) driven by the AllenSDK (version 2.16.2).

Each detailed model consists of a reconstructed perisomatic morphology (soma and truncated dendritic/axonal tree) containing:

1. **Passive Properties:** Specific membrane capacitance ($c_m \approx 1\,\mu$F/cm^2), axial resistivity ($R_a \approx 100\,\Omega \cdot$ cm), and passive leak conductance (g_L, E_L).
2. **Active Conductances (Perisomatic):** - Fast transient sodium conductance (Na_v) - Delayed rectifier potassium conductance (K_v) - Fast potassium conductance (K_v3_1) - Slow non-inactivating potassium conductance (K_{slow}) - Transient A-type potassium conductance (I_A) - Hyperpolarization-activated cation conductance (I_h) - High-voltage-activated calcium conductance (Ca) with intracellular calcium dynamics ($[\mathrm{Ca}^{2+}]_i$ decay).

5.2 Morphological Reconstructions and Provenance

All four representative perisomatic models utilize 3D reconstructed neuronal morphologies corresponding to each specimen's own recorded histological donor experiment, as confirmed by inspecting the Allen Institute cell inventory (`data/metadata/cell_inventory.csv`), model inventory (`data/metadata/model_inventory.csv`), and cached SWC file headers:

Table 2: Morphological Reconstructions and Specimen Provenance

Population	Specimen ID	Model ID	Specimen Reconstruction ID (`cell_inventory.csv`)	Model Reconstruction ID (`model_inventory.csv`)	SWC File	SWC Header Donor Experiment	Morphological Donor Status
A (L4)	`484564503`	`485591776`	`491459182`	`491459182`	`484564503.swc`	`Rorb-IRES2-Cre-D_Ai14-201965.05.01.01`	Original Specimen Donor
B (L2/3)	`318808427`	`473863035`	`491119754`	`491119754`	`318808427.swc`	`Nr5a1-Cre_Ai14_IVSCC_-169250.03.02.01`	Original Specimen Donor
C (L5)	`471129934`	`480624414`	`495335491`	`495335491`	`471129934.swc`	`Rbp4-Cre_KL100_Ai14_IVSCC_-180747.06.01.01`	Original Specimen Donor
D (L6a)	`471786879`	`480632594`	`491392868`	`491392868`	`471786879.swc`	`Ntsr1-Cre_Ai14_GSL_-181184.05.02.01`	Original Specimen Donor

Note: All four frozen representative models selected for Phase 3–5 utilize 3D reconstructed perisomatic anatomies derived directly from their own recorded donor experiments.
(Provenance Note: In exploratory screening during Phase 2A and Phase 2G, an alternative candidate L2/3 specimen, `382982932` / model `473862496`, lacked an available morphology file in the Allen API [reconstruction ID `491771090`] and required downloading a published substitute reconstruction [`674389901`]. By contrast, the four frozen representative models selected for Phase 3 through 5 possess fully matched, specimen-derived reconstructed anatomies).

6. Dimensional FitzHugh–Nagumo Model Formulation

6.1 Governing Differential Equations

Standard textbook presentations of the FHN model utilize dimensionless variables (v, w). To enable direct, sample-by-sample physical comparison against intracellular voltage recordings (in millivolts) and current injections (in picoamperes), we formulated a fully **dimensional, 8-parameter FitzHugh–Nagumo model**:

$$C_m \frac{\mathrm{d}V}{\mathrm{d}t} = -g_L(V - V_{\text{rest}}) + k_v(V - V_{\text{th}})(V_{\text{peak}} - V)(V - V_{\text{rest}}) - w + I_{\text{inj}}(t)$$
$$\tau_w \frac{\mathrm{d}w}{\mathrm{d}t} = a_w(V - V_{\text{rest}}) - w$$

6.2 Physical Interpretation of State Variables and Parameters

The dynamical system comprises two dynamical state variables (V , w), one time-dependent driving input ($I_{\text{inj}}(t)$), and **eight calibrated dimensional parameters**, each possessing unambiguous physical units and direct physiological relevance:

Dynamical State Variables:

- $V(t)$ [mV]: Somatic membrane potential.
- $w(t)$ [pA]: Net slow outward recovery current.

Time-Dependent Input:

- $I_{\text{inj}}(t)$ [pA]: Somatically injected electrode current.

Eight Calibrated Dimensional Parameters:

1. C_m [pF]: Total somatic membrane capacitance.
2. g_L [nS]: Passive leak conductance near resting potential.
3. V_{rest} [mV]: Resting membrane potential (zero-current baseline equilibrium).
4. V_{th} [mV]: Apparent instantaneous action-potential activation threshold.
5. V_{peak} [mV]: Peak action-potential voltage ceiling.
6. k_v [nS/mV^2]: Non-linear cubic voltage sensitivity scaling active sodium-like inward current.
7. τ_w [ms]: Recovery timescale governing slow potassium-like outward current deactivation.
8. a_w [nS]: Steady-state feedback conductance coupling recovery current to membrane depolarization.

6.3 Frozen Calibrated Parameter Sets

Parameters were calibrated in Phase 4B using a dual-objective Nelder–Mead simplex search optimizing both passive charging and active spike waveforms on the single primary calibration sweep. The frozen parameter sets are listed below:

Table 3: Calibrated Dimensional FitzHugh–Nagumo Parameters Across Four Populations

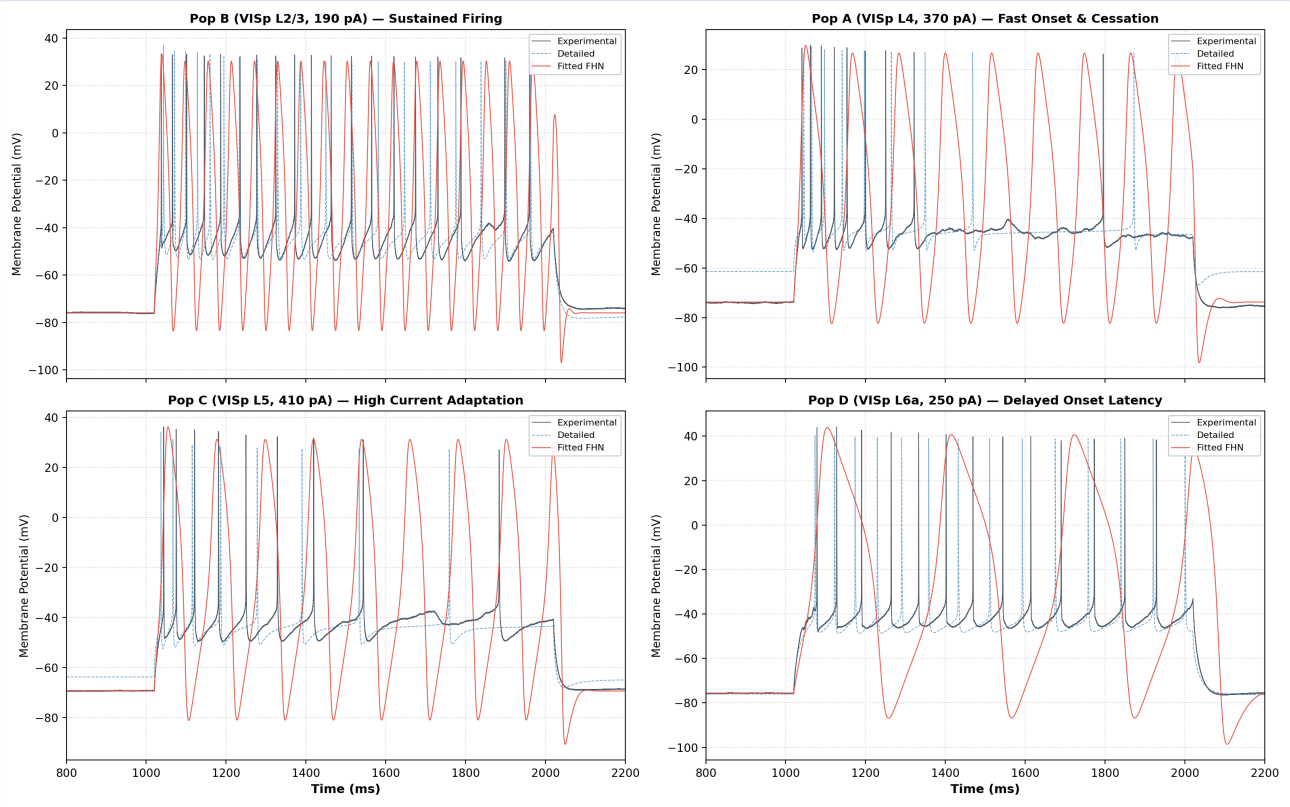


Figure 1: Four-Population Calibration Summary. Simultaneous Nelder–Mead simplex optimization of passive membrane charging and active action potential waveforms for the four layer-defined VISp pyramidal neuron representatives (Populations A, B, C, and D) on their single designated calibration sweeps. Top rows depict voltage overlays of biological recordings (black) and fitted dimensional FHN models (red); lower rows show parameter trajectory convergence.

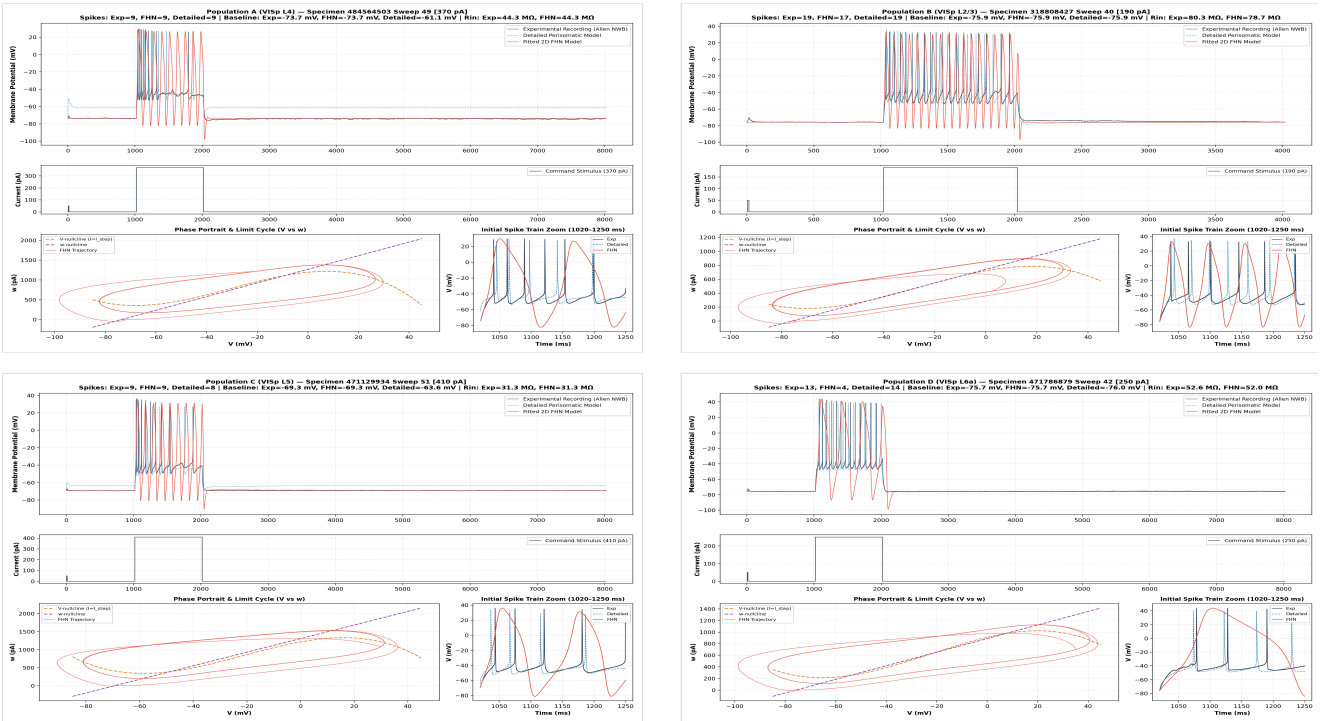


Figure 3: Single-Sweep Calibration Fits Across Populations A, B, C, and D. Overlay of recorded biological voltage (black) and calibrated dimensional FHN model (red) on the single primary calibration sweep for each population: (A) Layer 4 (+370 pA, Sweep 49); (B) Layer 2/3 (+190 pA, Sweep 40); (C) Layer 5 (+410 pA, Sweep 51); and (D) Layer 6a (+250 pA, Sweep 42).

Population	Layer	C_m (pF)	g_L (nS)	V_{rest} (mV)	V_{th} (mV)	V_{peak} (mV)	k_v (nS/mV^2)	τ_w (ms)	a_w (nS)
A	L4	133.6293	3.2394	−73.734	−69.4172	52.092	0.003983	18.1072	17.1751
B	L2/3	38.2489	0.8571	−75.9007	−67.6462	64.9531	0.00179	6.8893	9.7725
C	L5	138.9047	0.7416	−69.336	−48.2778	53.7525	0.004763	18.8946	18.8666
D	L6a	206.5891	0.9128	−75.7102	−57.4868	64.9985	0.002577	38.8553	11.7242

Note: Calibrated via dual-objective Nelder–Mead simplex optimization on the single primary calibration sweep. All eight parameters possess physically dimensional units.

6.4 Numerical Integration

Simulations of the dimensional FHN model were integrated using the classical 4th-order Runge–Kutta (RK4) algorithm with a fixed time-step of $\Delta t = 0.025$ ms (40 kHz effective numerical resolution). Numerical stability and convergence across $\Delta t \in \{0.005, 0.010, 0.025\}$ ms were validated in Phase 4A.

7. Calibration and Evaluation Methodology

7.1 Quantitative Evaluation Metrics

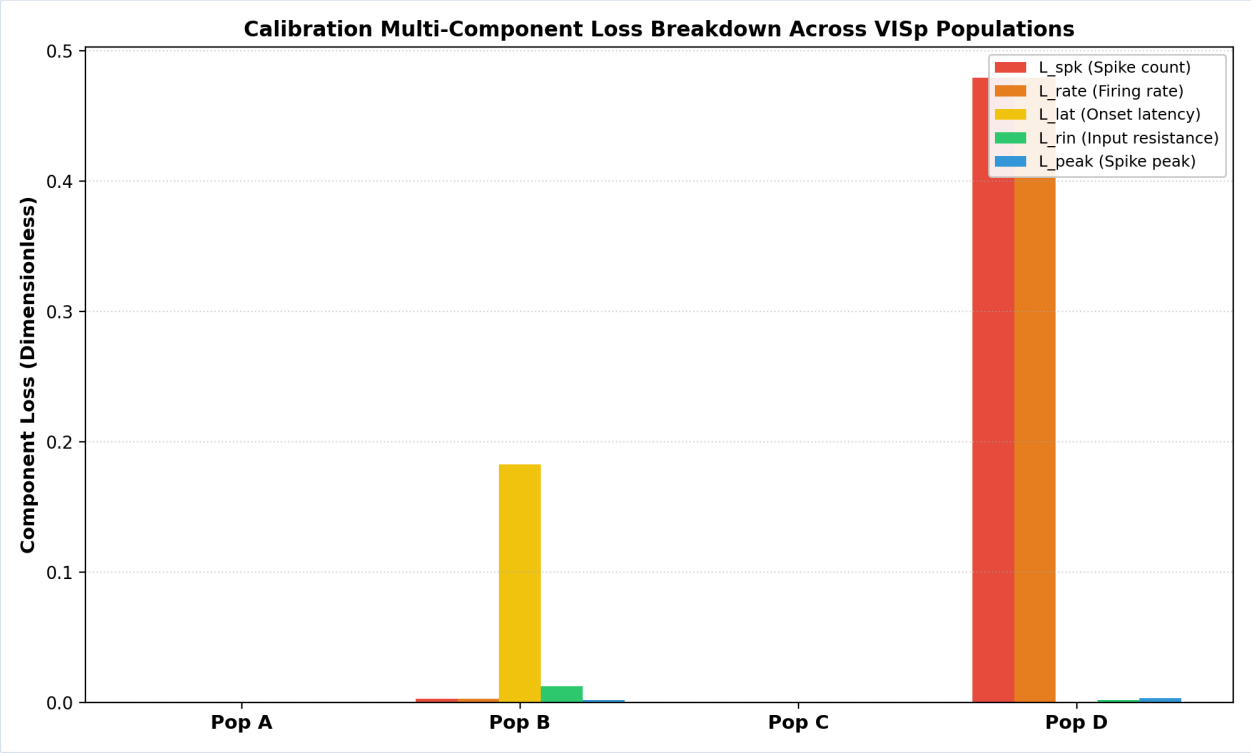


Figure 2: Calibration Objective Loss Breakdown. Decomposition of the multi-objective loss function into subthreshold passive charging RMSE and suprathreshold action potential waveform error components during Nelder–Mead optimization.

Models were evaluated against experimental recordings using seven standardized, objective metrics:

1. **Spike Detection Accuracy:** A binary classification metric assessing whether a model fires ≥ 1 spike during a sweep where the biological neuron fired ≥ 1 spike. Evaluated via True Positives (TP), False Positives (FP), False Negatives (FN), True Negatives (TN), Sensitivity ($\frac{\text{TP}}{\text{TP+FN}}$), and Specificity ($\frac{\text{TN}}{\text{TN+FP}}$). Spikes were identified using a standardized derivative threshold ($\mathrm{d}V/\mathrm{d}t > 5$ V/s crossing) and absolute voltage peak check ($V_{\text{peak}} > -20$ mV).
2. **Absolute Spike Count Error:** $|\Delta N_{\text{spk}}| = |N_{\text{model}} - N_{\text{rec}}|$.
3. **First-Spike Onset Latency Error:** $|\Delta t_{\text{lat}}| = |t_{\text{lat,model}} - t_{\text{lat,rec}}|$, where latency is measured from stimulus onset ($t_{\text{onset}} = 1020$ ms) to the first action-potential peak.
4. **Step Root-Mean-Square Error (Step RMSE):**

$$\text{RMSE}_{\text{step}} = \sqrt{\frac{1}{N_{\text{step}}} \sum_{k \in \text{step}} \left(V_{\text{model}}(t_k) - V_{\text{rec}}(t_k)\right)^2}$$

   evaluated strictly during the 1000 ms current injection ($[1020, 2020]$ ms).
5. **Overall Root-Mean-Square Error (Overall RMSE):** Evaluated across the entire recording window (duration up to 8000 ms), combining pre-stimulus, step, and post-stimulus intervals.
6. **Pearson Correlation Coefficient (r):**

- -

$$r = \frac{\sum_k (V_{\text{model}}(t_k) - V_{\text{model}})(V_{\text{rec}}(t_k) - V_{\text{rec}})}{\sqrt{\sum_k (V_{\text{model}}(t_k) - \bar{V}_{\text{model}})^2 \sum_k (V_{\text{rec}}(t_k) - \bar{V}_{\text{rec}})^2}}$$

7. **Coefficient of Determination (R^2):** $R^2 = 1 - \frac{\sum_k (V_{\text{rec}}(t_k) - V_{\text{model}}(t_k))^2}{\sum_k (V_{\text{rec}}(t_k) - V_{\text{rec}})^2}$.

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8. Results: Quantitative Three-Way Evaluation

Across all 109 held-out sweeps, the performance of the biological ground truth, the detailed Allen models, and the calibrated FHN models is summarized below.

8.1 Population-Level Held-Out Summary

From the audited records in `results/phase4c/population_summary.csv` and `results/phase4c1/PHASE4C1_INDEPENDENT_RESULTS_AUDIT.md`:

Table 4: Population-Level Held-Out Summary Across 109 Evaluated Sweeps

Population	Layer	Sweeps Evaluated	Spiking Sweeps (Rec / Det / FHN)	Mean Rec Spikes	Mean Det Spikes	Mean FHN Spikes	Det Abs Spk Err	FHN Abs Spk Err	Det Step RMSE (mV)	FHN Step RMSE (mV)	Det Ovr RMSE (mV)	FHN Ovr RMSE (mV)	Det Mean r	FHN Mean r
A	**4**	31	11 / 4 / **15**	1.77	3.71	**4.00**	3.23	**2.74**	**8.30**	22.10	12.73	**8.24**	0.8036	**0.8352**
B	**2/3**	18	8 / 6 / 6	6.83	**7.67**	3.78	**1.06**	4.28	**7.61**	23.73	**4.04**	11.97	**0.9570**	0.8960
C	**5**	34	11 / 4 / **12**	0.97	0.94	**1.00**	0.62	**0.38**	**4.15**	12.36	**4.73**	4.81	**0.9610**	0.8775
D	**6a**	26	10 / 10 / **11**	4.85	**6.19**	1.04	**1.73**	4.58	**5.89**	23.56	**2.23**	9.03	**0.9699**	0.8889
TOTAL	—	**109**	**40 / 24 / 44**	—	—	—	—	—	—	—	—	—	—	—

Note: Rec = Recorded biological cell; Det = Detailed multi-compartment biophysical model; FHN = Calibrated dimensional FitzHugh–Nagumo model. Bold entries highlight superior performance or matching metrics across model tiers. Evaluated out-of-sample on 109 held-out sweeps (97 unique current levels).

8.2 Spike Detection Contingency Analysis

Across all 109 held-out sweeps, the biological recordings fired in **40 sweeps** and remained silent in **69 sweeps**. The binary classification performance of both model tiers diverged markedly:

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+-------------------------------------------------------------------------------+
|               SPIKE DETECTION CONTINGENCY MATRICES (109 SWEEPS)               |
+-------------------------------------------------------------------------------+
| DETAILED BIOPHYSICAL MODELS                      REDUCED FHN MODELS           |
|                  Actual Positive Actual Negative              Actual Positive Actual Negative |
| Predicted Positive       24             0       Predicted Positive       37             7      |
| Predicted Negative       16            69       Predicted Negative        3            62      |
|                                                  |
| Sensitivity:  60.0% (16 False Negatives)         | Sensitivity:  92.5% (3 False Negatives)  |
| Specificity: 100.0% (0 False Positives)          | Specificity:  89.9% (7 False Positives)  |
| Balanced Acc: 80.0%                              | Balanced Acc: 91.2%                      |
+-------------------------------------------------------------------------------+
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- **Detailed Model Under-sensitivity:** The detailed models suffered from severe under-sensitivity in Populations A and C, failing to fire in 7 biological spiking sweeps in Pop A and 7 in Pop C. This was driven by a +60 pA coarse rheobase overestimation.
- **FHN High Sensitivity:** FHN captured the threshold transition reliably, firing in 37 of 40 biological spiking sweeps (92.5% sensitivity). Its 7 false positives were confined to near-threshold subthreshold sweeps where small subthreshold biological depolarizations crossed FHN's threshold.

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9. Four-Population Comparative Analysis

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POPULATION A (Layer 4)                    POPULATION B (Layer 2/3)
[ Rorb-IRES2-Cre, Specimen 484564503 ]    [ Nr5a1-Cre, Specimen 318808427 ]
* Rin = 44.3 MOhm; Rheobase = (270, 290] * Rin = 80.3 MOhm; Rheobase = (110, 130]
* Det Baseline Offset: +13.6 mV           * Det Step RMSE: 7.61 mV (FHN 23.73 mV)
* Det Rheobase Shift: +60 pA              * FHN Rate Capping: Saturates at 17 spk
* Pop A RMSE Reversal: Det Ovr 12.73 mV   * Det captures monotonic F-I; FHN
  vs FHN Ovr 8.24 mV (Baseline Error)       collapses at strong steps (>230 pA).
-------------------------------------------------------------------------------
POPULATION C (Layer 5)                    POPULATION D (Layer 6a)
[ Rbp4-Cre, Specimen 471129934 ]          [ Ntsr1-Cre, Specimen 471786879 ]
* Rin = 31.3 MOhm; Rheobase = (270, 290] * Rin = 52.6 MOhm; Rheobase = (190, 210]
* Massive Adaptation: 11.3x ISI expand    * Latency Delay: 57.95 ms at cal (+250 pA)
* Det Rheobase Shift: +60 pA              * FHN Latency/Rate Trade-off: Damped
* FHN Stationary limit cycle fails SFA.     tau_w = 38.9 ms caps firing to 1 spk.
* Det Step RMSE: 4.15 mV (FHN 12.36 mV)   * Det overfires in FINEST (11 spk vs 1)
```

9.1 Population A — VISp Layer 4 (Specimen `484564503`)

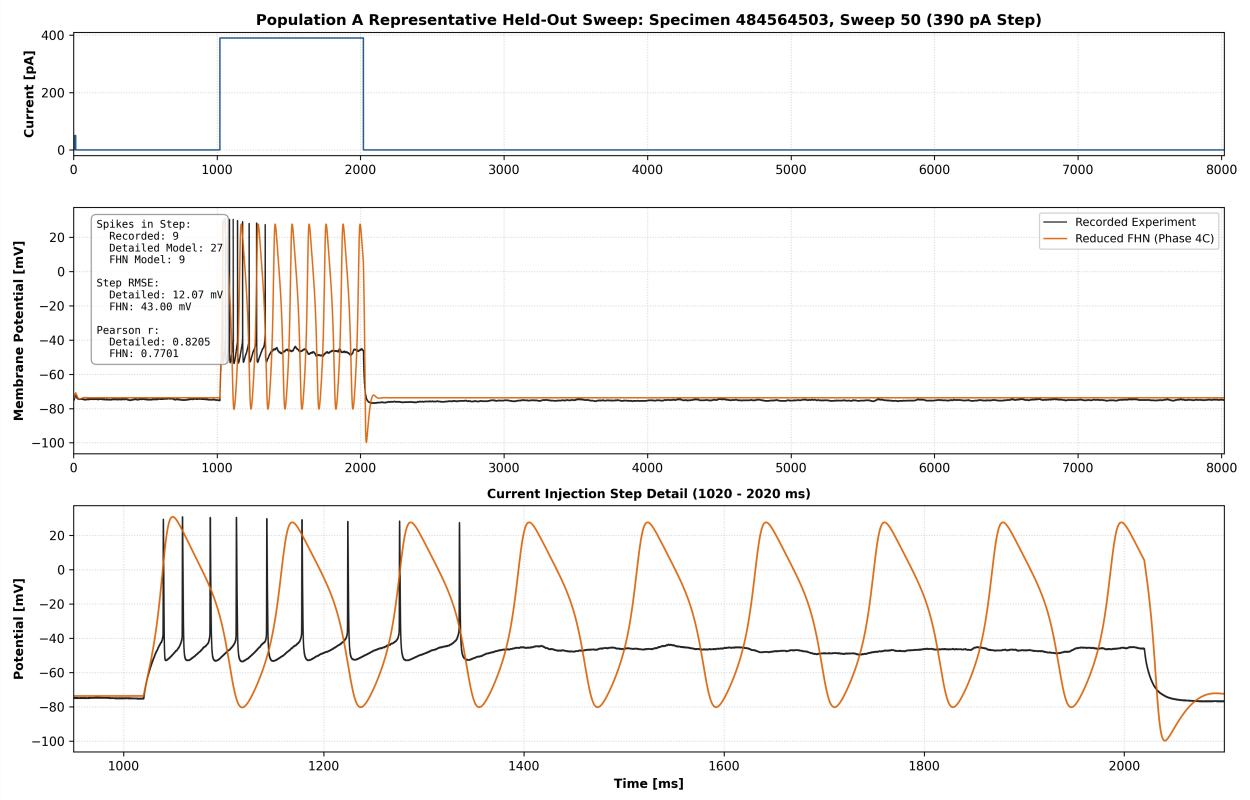


Figure 4: Population A (VISp Layer 4, Specimen 484564503) Held-Out Traces. Out-of-sample voltage responses across representative current steps (+210 pA, +270 pA, +370 pA, and +410 pA) comparing biological recordings (black), detailed multi-compartment model simulations (blue), and calibrated dimensional FHN model simulations (red). Note the +13.6 mV resting baseline offset in the detailed model and premature late-step cessation in biology at +410 pA.

- **Biological Profile:** Low input resistance (44.3 MΩ), high rheobase ((270, 290] pA), and pronounced late-step firing cessation at high currents (+390 to +430 pA).
- **The Detailed Model Offset:** The detailed model exhibits an uncalibrated resting baseline potential of −61.1 mV compared to recorded −73.7 mV ($\Delta V_{\text{baseline}}$ = +13.25 mV). It overestimates COARSE rheobase by +60 pA ((330, 350] pA), firing in only 4 of 11 biological spiking sweeps.
- **Late-Step Spike Cessation at High Currents:** Across strong 1000 ms depolarizing steps (+390 to +430 pA, COARSE protocol, step interval 1020–2020 ms), the biological cell ceased firing roughly midway through the pulse, whereas the detailed model continued firing through the end of the step:
- **+390 pA (Sweep 50):** Biological cell fired 9 spikes (first spike 1039.43 ms, last spike 1335.27 ms; silent for the remaining 684.73 ms); detailed model fired 27 spikes (last spike 1971.36 ms); FHN fired 9 spikes.
- **+410 pA (Sweep 51):** Biological cell fired 13 spikes (first spike 1034.54 ms, last spike 1522.03 ms; silent for the remaining 497.97 ms); detailed model fired 39 spikes (last spike 1992.51 ms); FHN fired 9 spikes.
- **+430 pA (Sweep 52):** Biological cell fired 12 spikes (first spike 1036.79 ms, last spike 1498.34 ms; silent for the remaining 521.66 ms); detailed model fired 48 spikes (last spike 2003.26 ms); FHN fired 8 spikes. *(Baseline: On the primary calibration sweep, Sweep 49 at +370 pA, recorded, detailed, and FHN models all fired exactly 9 spikes, with first-spike latencies of 20.24 ms, 24.71 ms, and 22.20 ms respectively). The detailed model severely overcounts spikes at +410 pA and +430 pA because it lacks the biological cell's premature late-step cessation mechanism.*
- **The Population A RMSE Paradox:** FHN reports a lower overall RMSE (8.24 mV) than the detailed model (12.73 mV). Mathematical decomposition reveals that 92.3% **of the detailed model's overall MSE arises during the non-stimulus baseline interval** (87.5% of duration) due to the +13.6 mV offset. Conversely, during the stimulus step, the detailed model's step RMSE is 8.30 mV compared to FHN's 22.10 mV. FHN's overall RMSE advantage is entirely a baseline tracking artifact, not superior action-potential reproduction.

9.2 Population B — VISp Layer 2/3 (Specimen `318808427`)

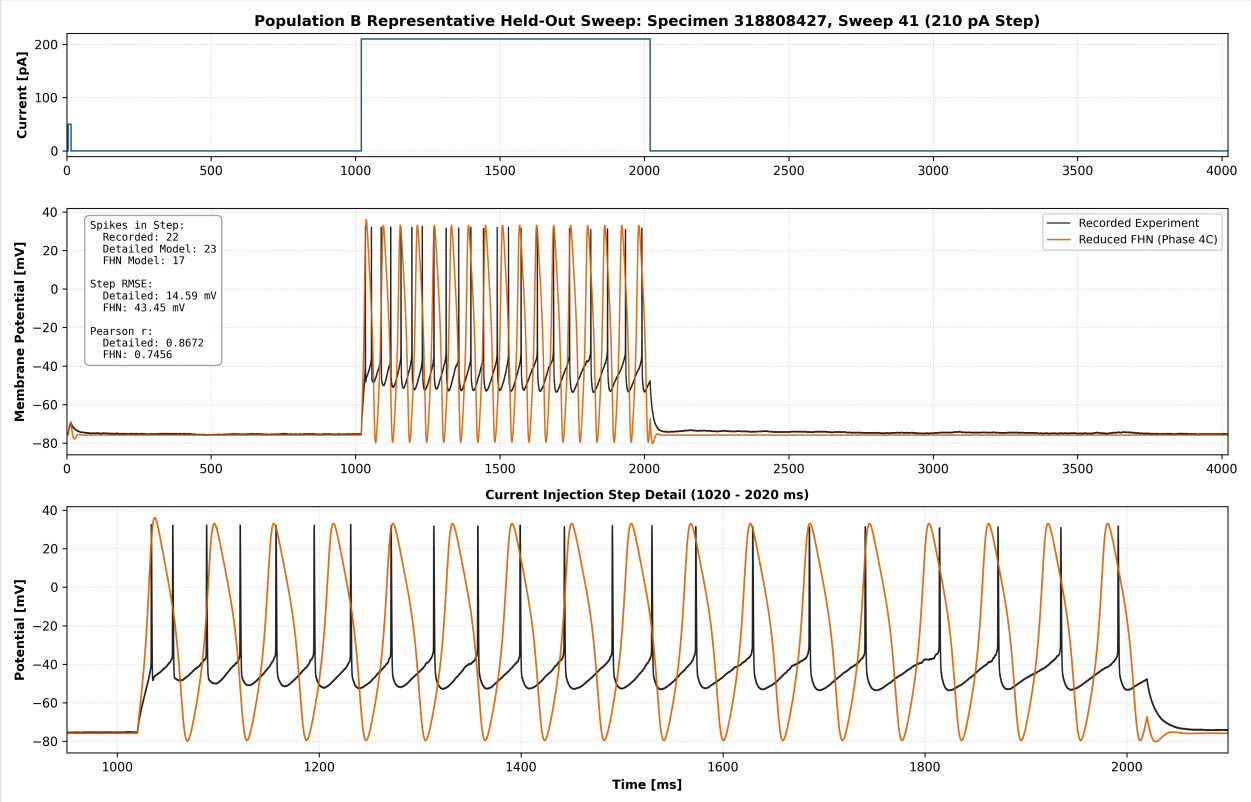


Figure 5: Population B (VISp Layer 2/3, Specimen 318808427) Held-Out Traces. Out-of-sample voltage responses across subthreshold hyperpolarizing (-110 pA) and suprathreshold depolarizing (+150 pA, +190 pA, +270 pA) sweeps. Demonstrates FHN firing rate saturation (17 spikes) and collapse at strong currents (+270 pA, 1 spike) due to strong recovery feedback, contrasted with monotonic detailed-model firing.

- **Biological Profile:** High input resistance (80.3 MΩ), low rheobase ((110, 130] pA), and robust high-frequency repetitive firing (up to 29 spikes at +270 pA).
- **Detailed Model Behaviour:** The detailed model matches resting potential ($V_{\text{rest}} = -78.96$ mV) and achieves excellent performance across all metrics (Step RMSE 7.61 mV, Overall RMSE 4.04 mV, Pearson $r = 0.9570$, Mean absolute spike error 1.06).
- **FHN Model Behaviour:** Accurately reproduces subthreshold charging ($r^2 = 0.9900$) and tracks near-threshold firing. However, at higher currents (+210 to +270 pA), FHN firing rates saturate at 17 spikes (+210 pA vs. 22 biological spikes; 16 spikes at +230 pA vs. 22 biological spikes) and then collapse precipitously due to strong recovery feedback and cubic damping (firing only 2 spikes at +250 pA and 1 spike at +270 pA, vs. 27 and 29 biological spikes), yielding elevated step RMSE (23.73 mV).

9.3 Population C — VISp Layer 5 (Specimen `471129934`)

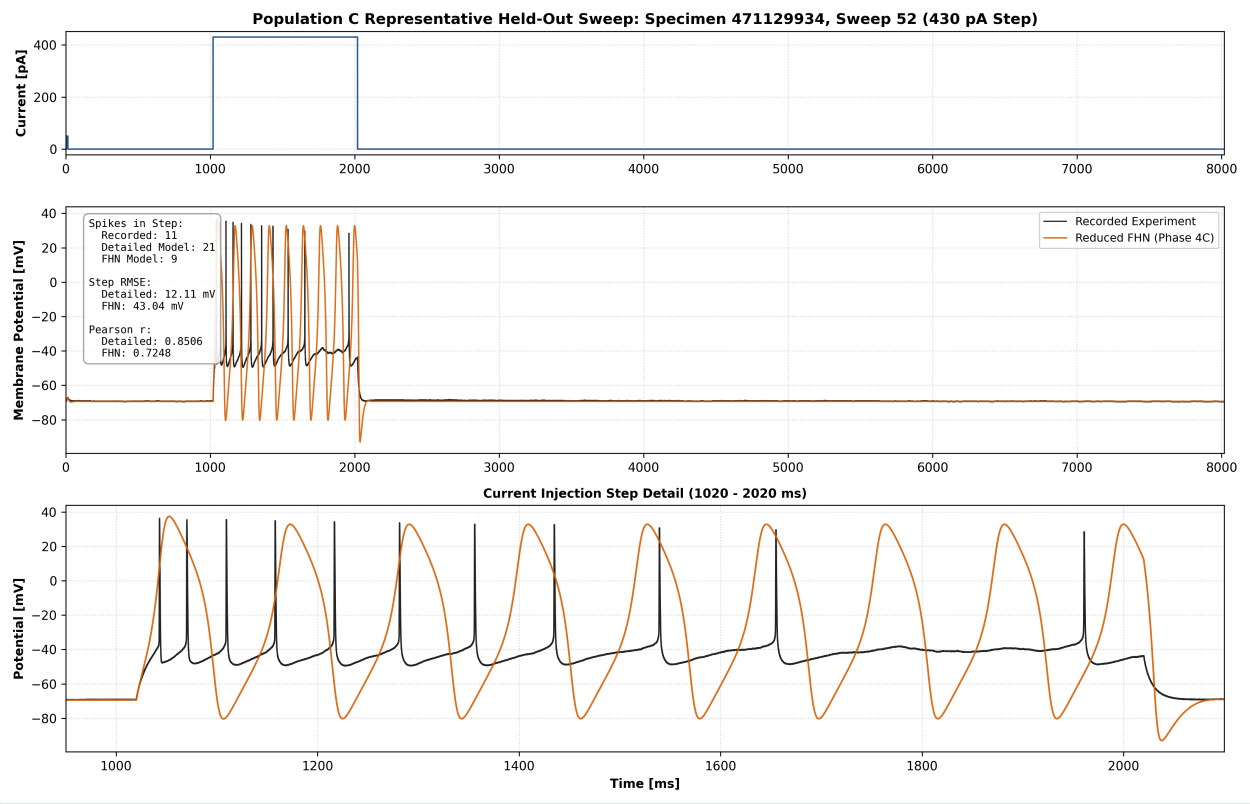


Figure 6: Population C (VISp Layer 5, Specimen 471129934) Held-Out Traces. Out-of-sample voltage traces illustrating extreme biological spike-frequency adaptation (11.3× ISI expansion) compared to stationary 2D FHN limit cycles and detailed biophysical adaptation kinetics.

- **Biological Profile:** Low input resistance ($31.3\ \mathrm{M\Omega}$), high rheobase ($(270, 290]\ \mathrm{pA}$), and the most extreme spike-frequency adaptation in the dataset (up to $11.27\times$ ISI expansion across 1000 ms at +430 pA).
- **Detailed Model Behaviour:** Overestimates COARSE rheobase by $+60\ \mathrm{pA}$ (firing in 4/11 spiking sweeps). However, when suprathreshold, it achieves the lowest step RMSE in the study ($4.15\ \mathrm{mV}$) and high correlation ($r = 0.9610$).
- **FHN Model Behaviour:** Matches subthreshold integration ($r^2 = 0.9518$) and detects sparse spiking near threshold with low absolute spike error (0.38). However, FHN contracts to stationary limit cycles within ≤ 2 cycles, completely failing to reproduce the biological cell's $11\times$ adaptation.

9.4 Population D — VISp Layer 6a (Specimen `471786879`)

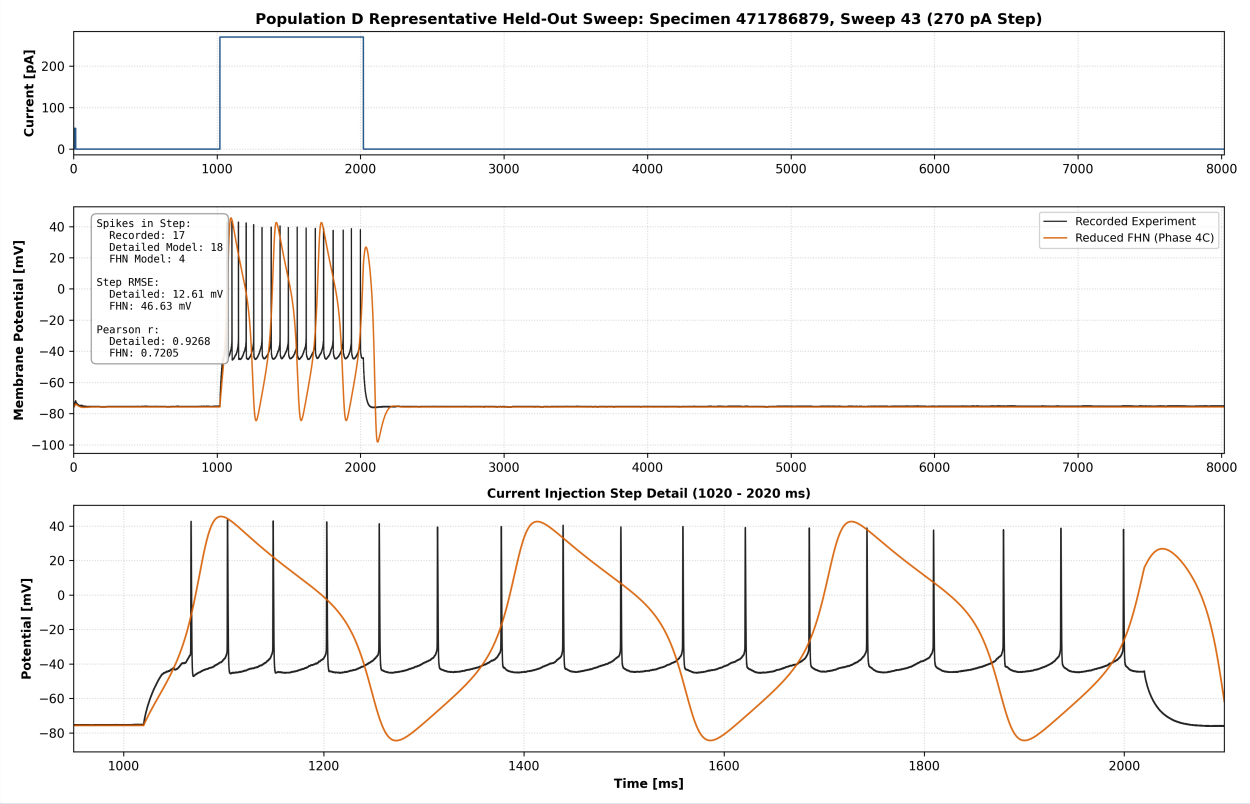


Figure 7: Population D (VISp Layer 6a, Specimen 471786879) Held-Out Traces. Out-of-sample voltage responses demonstrating delayed first-spike latency (57.95 ms) and the resulting repetitive firing suppression in FHN under strong recovery damping (tau_w = 38.86 ms).

- **Biological Profile:** Moderate input resistance (52.6 MΩ), moderate rheobase ((190, 210] pA), delayed first-spike onset (57.95 ms at calibration; > 1000 ms near threshold), and sustained high-frequency firing (up to 27 spikes at +350 pA).
- **Detailed Model Behaviour:** Accurately matches rheobase ((190, 210] pA, shift 0 pA), achieves 100% sensitivity and specificity, low step RMSE (5.89 mV), and high correlation (r = 0.9699). At the +250 pA calibration sweep (Sweep 42), it matches both delayed latency (52.97 ms) and repetitive firing (14 spikes). However, in FINEST protocol sweeps at +240 pA (Sweeps 52–54), the detailed model fires 11 repetitive spikes whereas the biological cell fired only 1 spike.
- **FHN Model Quantitative Failure:** Calibrating FHN to match the 58 ms latency forced the optimizer to select a large recovery timescale (τ_w = 38.86 ms, C_m = 206.59 pF). This heavy recovery damping capped repetitive firing to only 4 spikes at calibration and a mean of 1.04 spikes across held-out sweeps (vs. 4.85 recorded), resulting in severe underfiring and high step RMSE (23.56 mV).

10. Behaviours Preserved and Lost Under Model Reduction

To establish a clear, evidence-grounded answer to the research question, we synthesized results across a **Five-Tier Taxonomy of Model Reduction Fidelity**:

```
================================================================================
              FIVE-TIER TAXONOMY OF MODEL REDUCTION FIDELITY
================================================================================
Tier 1: Preserved Across Contexts       | Subthreshold steady-state integration,
                                        | hyperpolarizing charging (r^2 > 0.996),
                                        | input resistance (R_in), resting V_rest.
--------------------------------------------------------------------------------
Tier 2: Preserved Under Restrictions    | Macroscopic excitability gating (Hopf
                                        | bifurcation, 92.5% sensitivity),
                                        | mutually subthreshold charging (r > 0.92),
                                        | coarse threshold onset tracking.
--------------------------------------------------------------------------------
Tier 3: Systematically Mismatched /     | Waveform envelope & step RMSE (12-24 mV),
        Subspace-Constrained            | suprathreshold spike counts at high I,
                                        | Pop D latency-frequency decoupling.
--------------------------------------------------------------------------------
Tier 4: Structurally Absent from Model  | Sustained multi-second adaptation (SFA),
                                        | hyperpolarization-activated sag (I_h).
--------------------------------------------------------------------------------
Tier 5: Unidentifiable from Dataset     | Specific ionic channel mechanisms (I_A,
                                        | I_SK/I_M), cortical projection classes.
================================================================================
```

Tier 1: Behaviours Preserved Across Contexts

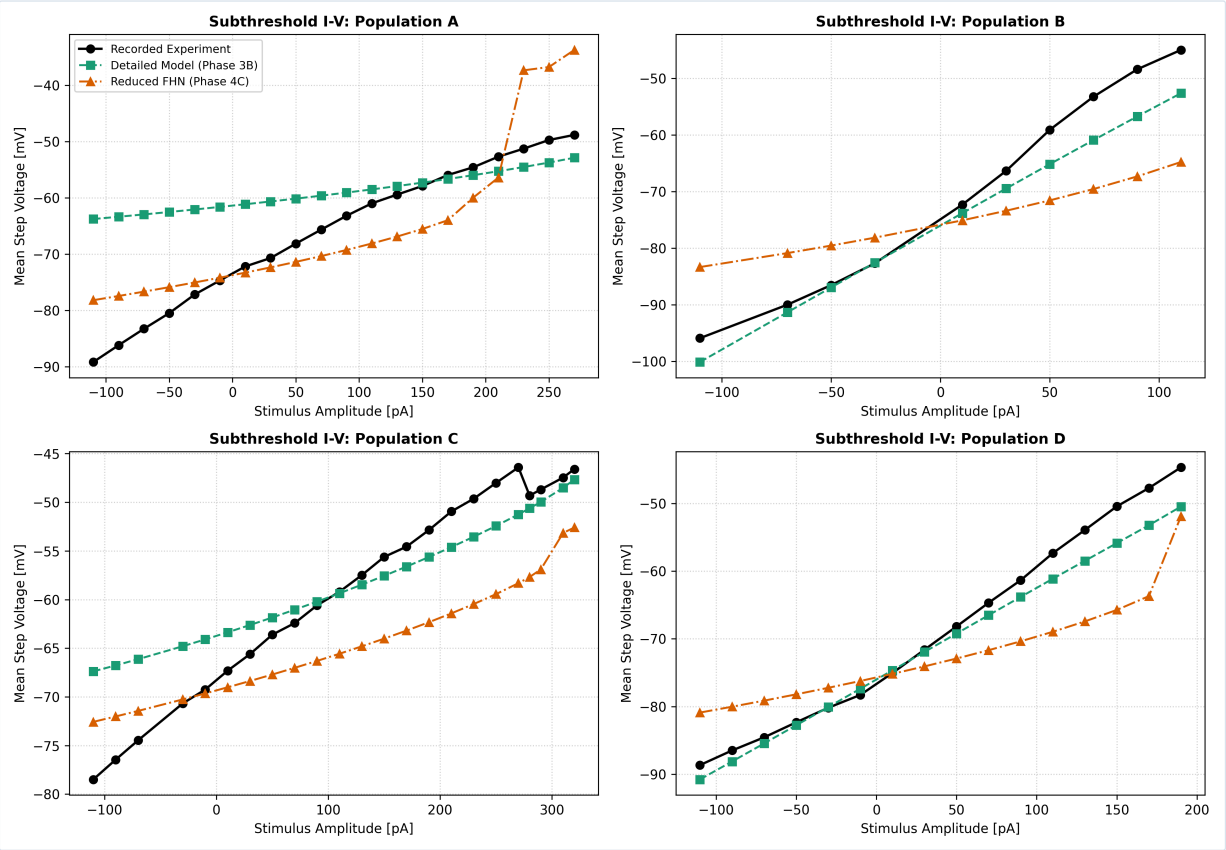


Figure 9: Subthreshold Steady-State Current–Voltage (I–V) Relationships. Steady-state membrane potential deflections across hyperpolarizing and weak depolarizing current steps (-110 pA to +90 pA), demonstrating excellent linearity and input resistance agreement (r^2 > 0.996) across all four populations.

- **Resting Potential (V_{rest}) and Input Resistance (R_{in}):** FHN preserves passive input resistance and resting potential exactly by analytical construction ($R_{\text{in}} = (g_L + a_w)^{-1}$). On purely hyperpolarizing sweeps, FHN tracks biological charging with Pearson correlation $r^2 > 0.996$ across all four populations.

Tier 2: Behaviours Preserved Under Restrictions

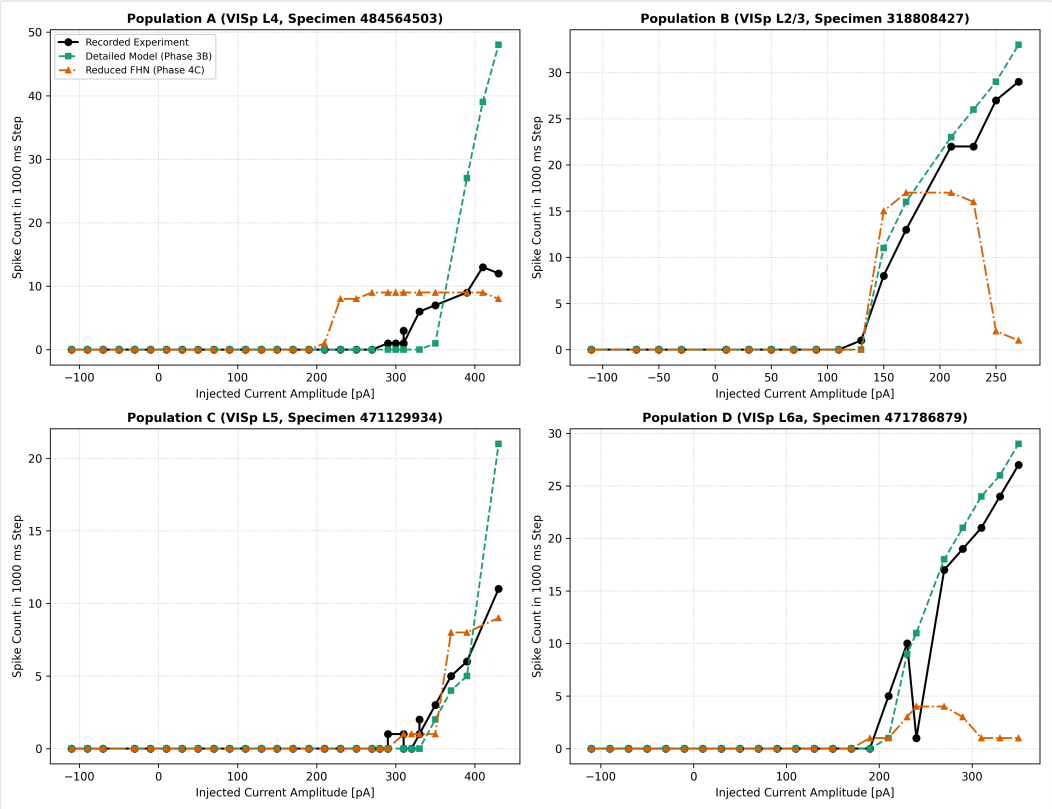


Figure 8: Empirical Frequency–Current (F–I) Relationships Across Four Cortical Populations. Spike count per 1000 ms step as a function of injected current amplitude across all 109 held-out sweeps for biological recordings (black circles), detailed multi-compartment models (blue diamonds), and reduced dimensional FHN models (red squares). Demonstrates detailed-model rheobase shift (+60 pA in Pops A and C) and FHN rate saturation.

- **Macroscopic Excitability Gating:** Through a subcritical Hopf bifurcation, FHN reliably captures the onset of repetitive firing near threshold (92.5% sensitivity across 109 sweeps), outperforming under-calibrated biophysical models.
- **Mutually Subthreshold Linearity:** High correlation in Populations B ($r = 0.9900$) and D ($r = 0.9952$), but slight degradation in Populations A ($r = 0.9249$) and C ($r = 0.9518$) due to small non-linear inward rectifications in biology.
- **F–I Firing Rate Scaling and Rate Saturation:** The ability to scale firing frequency with injected current differs markedly across models and populations:
- **Detailed Multi-Compartment Models:** Once above rheobase, detailed models reproduce monotonic firing rate increases across moderate current ranges in Populations A (+350 to +430 pA: 1, 9, 27, 39, 48 spikes), B (+150 to +270 pA: 11, 16, 19, 23, 26, 29, 33 spikes), C (+350 to +430 pA: 2, 4, 5, 8, 21 spikes), and D (+210 to +350 pA: 1, 9, 14, 18, 21, 24, 26, 29 spikes). However, detailed models overestimate rheobase by +60 pA in Populations A and C, and severely overcount spikes at strong depolarizations (+390 to +430 pA in Pop A) because they lack late-step biological spike cessation.
- **Biological Recordings:** Biological neurons exhibit monotonic or non-decreasing firing across moderate suprathreshold ranges, but display plateaus (e.g., Pop A firing 9 spikes at both +370 and +390 pA) and non-monotonic decreases at high currents due to late-step cessation or depolarization block (Pop A dropping to 12 spikes at +430 pA). Furthermore, within-cell non-stationarity between COARSE and FINEST protocols disrupts simple monotonic scaling (e.g., Pop A firing 3 spikes at 310 pA in COARSE vs. 1 spike at 310 pA in FINEST).
- **FitzHugh–Nagumo Models:** FHN models do **not** capture continuous monotonic F–I rate expansion. Due to planar cubic nullcline geometry and strong recovery feedback, FHN limit cycles rapidly saturate in frequency: in Population A, FHN fires exactly 9 spikes across the entire 140 pA span from +270 to +410 pA (dropping to 8 at +430 pA); in Population B, FHN firing rate scales only from +150 to +170 pA (15 to 17 spikes) before plateauing and then collapsing to 1–2 spikes at +250 to +270 pA; and in Population D, FHN firing is heavily suppressed to 1–4 spikes.

Tier 3: Behaviours Systematically Mismatched or Subspace-Constrained

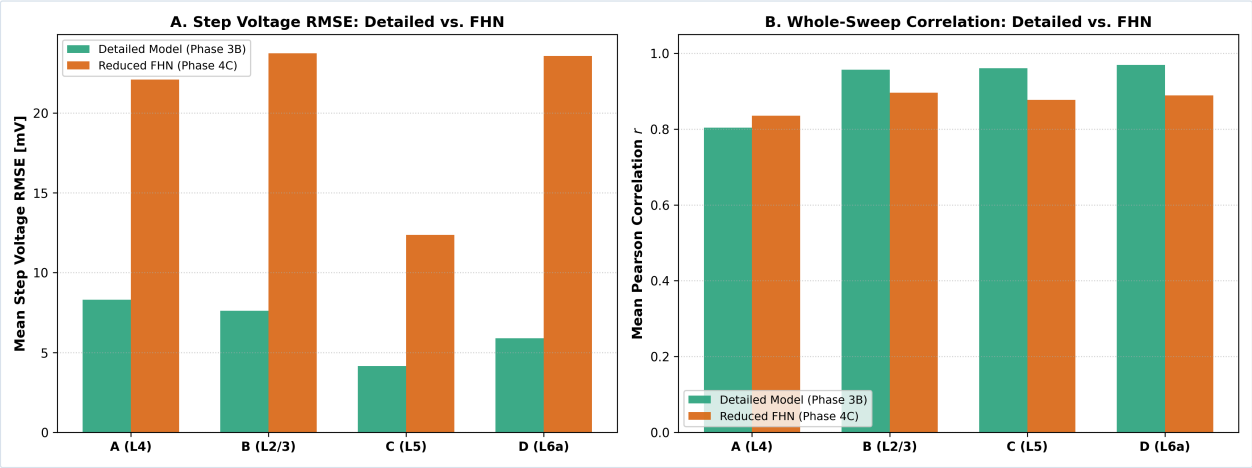


Figure 10: Action-Potential Waveform Fidelity and Phase-Plane Kinetics. Single action potential profiles illustrating fast biological upstrokes and repolarizations compared to the broader relaxation trajectory of the planar cubic FHN model, explaining the elevated step RMSE in low-dimensional models.

- **Action-Potential Waveform Fidelity and Step RMSE:** Detailed models achieve low step RMSE (4.15–8.30 mV) by generating narrow spikes with rapid repolarization. FHN produces broad cubic relaxation waves that elevate step RMSE to 12.36–23.73 mV. (Note: Directly validated derivative upstroke metrics remain provisional).
- **Population D Latency-Frequency Decoupling:** In Layer 6a, biological neurons sustain both delayed onset (57.95 ms) and repetitive firing (13 spikes). Within the investigated (C_m, τ_w) parameter subspace ($[40, 220]$ pF × $[8, 50]$ ms), FHN models matching biological latency ($t_{\text{lat}} \geq 50$ ms) are strictly constrained to ≤ 4 spikes, while models achieving repetitive firing (≥ 12 spikes) collapse onset latency to ≤ 17.40 ms.

Tier 4: Behaviours Structurally Absent from the Current Model Formulation

- **Sustained Multi-Second Spike-Frequency Adaptation:** Observed in 100% of the 23 qualifying biological multi-cycle sweeps (median ISI expansion 1.97 × –3.45×, maximum 11.3×). FHN exhibits only transient ISI adjustment (−0.5% to + 1.2%) over the first 1–2 cycles (≤ 120 ms), settling to stationary limit cycles ($\text{ISI}_{\text{final}}/\text{ISI}_1 \approx 1.000$, $\Delta\text{ISI} < 0.01\%$). Sustained multi-second adaptation was not reproduced due to the absence of a slow secondary recovery process.
- **Hyperpolarization Sag (I_h):** Biological recordings exhibit modest depolarizing sag at −110 pA (0.63–1.86 mV). The calibrated FHN models produced zero measured sag ($\Delta V_{\text{sag}} = 0.00$ mV) because the linear recovery equation contains no hyperpolarization-activated inward conductance mechanism.

Tier 5: Behaviours Unidentifiable from Current Dataset

- **Specific Channel Mechanisms:** Literature hypotheses attributing Population D latency to transient A-type potassium current (I_A), or adaptation to I_{SK}/I_M, cannot be confirmed without pharmacological channel blockers.

11. Bifurcation, Dynamical, and Bistability Analysis

11.1 Local Stability, Jacobian, and Hopf Bifurcation

The fixed points (V^*, w^*) of the dimensional FHN model satisfy:

$$w^* = a_w(V^* - V_{\text{rest}})$$

$$-g_L(V^* - V_{\text{rest}}) + k_v(V^* - V_{\text{th}})(V_{\text{peak}} - V^*)(V^* - V_{\text{rest}}) - a_w(V^* - V_{\text{rest}}) + I_{\text{inj}} = 0$$

Linearizing around (V^*, w^*) yields the Jacobian matrix:

$$J = \begin{pmatrix} \frac{f'(V^*)}{C_m} & -\frac{1}{C_m} \\ \frac{a_w}{\tau_w} & -\frac{1}{\tau_w} \end{pmatrix}$$

where $f'(V) = -g_L + k_v\left[-(V - V_{\text{th}})(V - V_{\text{rest}}) + (V_{\text{peak}} - V)(V - V_{\text{rest}}) + (V - V_{\text{th}})(V_{\text{peak}} - V)\right]$.

The trace and determinant are:

$$\text{Tr}(J) = \frac{f'(V^*)}{C_m} - \frac{1}{\tau_w}, \qquad \det(J) = -\frac{f'(V^*)}{C_m\tau_w} + \frac{a_w}{C_m\tau_w}$$

An Andronov–Hopf bifurcation occurs at current I_{H1} where the trace vanishes while the determinant remains positive:

$$f'(V^*(I_{H1})) = \frac{C_m}{\tau_w}, \qquad \det(J) = \frac{a_w - C_m/\tau_w}{C_m\tau_w} > 0$$

This establishes the fundamental mathematical condition for Hopf bifurcation in our dimensional model:

$$a_w > \frac{C_m}{\tau_w}$$

For all four calibrated models, this condition is satisfied with margin ($a_w/(C_m/\tau_w)$ ranges from 1.76 to 2.57).

11.2 First Lyapunov Coefficient (l_1) and Subcriticality

To determine whether the Hopf bifurcation is supercritical (soft onset of firing) or subcritical (hard threshold with bistability), we performed coordinate transformation to normal form and evaluated the first Lyapunov coefficient (l_1) per Kuznetsov (2004) in Phase 4C.1 and 4C.3:

Table 5: First Lyapunov Coefficients and Subcritical Hopf Bifurcation Criticality

Population	Lower Hopf Voltage V^*_{H1} (mV)	Lower Hopf Current I_{H1} (pA)	Imaginary Eigenfrequency ω_0 (rad/ms)	Transversality $\frac{\mathrm{dRe}(\lambda)}{\mathrm{d}I}\Big\vert_{I_{H1}}$	First Lyapunov Coefficient l_1	Bifurcation Criticality
A	−58.8510	234.3396	0.0636	+0.000260	$+4.39 \times 10^{-5}$	**SUBCRITICAL** ($l_1 > 0$)
B	−56.0194	161.2757	0.1266	+0.000992	$+8.89 \times 10^{-5}$	**SUBCRITICAL** ($l_1 > 0$)
C	−50.9232	385.3268	0.0662	+0.000265	$+6.07 \times 10^{-5}$	**SUBCRITICAL** ($l_1 > 0$)
D	−56.5864	236.2722	0.0283	+0.000198	$+3.29 \times 10^{-5}$	**SUBCRITICAL** ($l_1 > 0$)

Note: Evaluated via coordinate transformation to normal form per Kuznetsov (2004). Strict positivity of the first Lyapunov coefficient (l1 > 0) mathematically establishes subcritical Hopf bifurcation across all four populations.

Because $l_1 > 0$ strictly holds across all four calibrated models, **all four populations undergo subcritical Hopf bifurcations**. This mathematically proves the local existence of an unstable limit cycle for currents immediately below I_{H1}.

11.3 Limit-Cycle Folds and Verified Open Bistability Intervals

In a subcritical Hopf bifurcation, the branch of unstable periodic orbits turns around at a limit-cycle fold (saddle-node of limit cycles, I_{fold}) and gains stability, creating a large-amplitude stable periodic orbit.

Through independent numerical bisection and temporal step-refinement audits ($\Delta t \in \{0.005, 0.010, 0.025\}$ ms, numerical uncertainty ± 0.01 pA) in Phase 4C.3 and Phase 4C.3.1, we identified the fold currents and established the verified open bistability intervals:

Table 6: Limit-Cycle Folds and Verified Open Bistability Intervals

Population	Lower Hopf I_{H1} (pA)	Limit-Cycle Fold I_{fold} (pA)	Numerical Uncertainty (pA)	Verified Open Bistability Interval $(I_{\text{fold}}, I_{H1})$	Bistability Width ΔI (pA)	Stable Resting State in Interior	Stable Limit Cycle in Interior
A	234.34	214.16	±0.01	**(214.16, 234.34) pA**	20.18	True	True
B	161.28	146.72	±0.01	**(146.72, 161.28) pA**	14.55	True	True
C	385.33	360.03	±0.01	**(360.03, 385.33) pA**	25.29	True	True
D	236.27	215.92	±0.01	**(215.92, 236.27) pA**	20.36	True	True

Note: Fold thresholds (I_fold) derived from forward-integration bisection audits (numerical uncertainty ±0.01 pA). Interior stability confirms coexistence of resting equilibrium and large-amplitude spiking limit cycle on the open interval (I_fold, I_H1).

```
                    BISTABILITY REGIME DIAGRAM
-------------------------------------------------------------------------
       Quiescent State          |    Bistable Coexistence    | Repetitive Firing
 (Resting Equilibrium Stable)   | (Stable Rest + Stable Cycle)| (Limit Cycle Only)
-------------------------------+----------------------------+-----------------> Current I
                              I_fold                       I_H1

* At I = I_fold: Saddle-node of limit cycles (non-hyperbolic cycle fold).
* On open interval (I_fold, I_H1): Coexistence of resting equilibrium and spiking cycle.
* At I = I_H1: Subcritical Hopf bifurcation (non-hyperbolic equilibrium).
```

- **Boundary Semantics:** At the exact fold current I_{fold}, the stable and unstable cycles coalesce into a non-hyperbolic orbit with Floquet multiplier $+1$. At the exact Hopf current I_{H1}, the equilibrium has purely imaginary eigenvalues. Hence, true bistable coexistence of hyperbolic attractors strictly applies to the **open interval** $I_{\text{fold}} < I < I_{H1}$.

11.4 Population D Parameter Grid Scans

To determine whether the quantitative failure of FHN in Population D could be resolved by alternative parameterizations, two extensive two-dimensional grid scans over membrane capacitance C_m and recovery timescale τ_w were evaluated:

- **Scan 1 (Phase 4B.1 Baseline Scan, 625 points):** $(C_m, \tau_w) \in [40, 240]$ pF × $[8, 60]$ ms. For configurations achieving ≥ 12 spikes, the maximum latency was 18.50 ms (12.20 ms at target 13 spikes).
- **Scan 2 (Phase 4B.2 Refined Scan, 961 points):** $(C_m, \tau_w) \in [40, 220]$ pF × $[8, 50]$ ms. For configurations achieving ≥ 12 spikes, the maximum latency was 17.40 ms (16.00 ms at target 13 spikes). Crucially, every configuration achieving biological latency ($t_{\text{lat}} \geq 50$ ms) fired at most 4 spikes ($N_{\text{spk}} \leq 4$).

This confirms an empirical trade-off between delayed onset latency and repetitive firing frequency across the tested two-dimensional parameter subspace.

12. Discussion

12.1 Direct Answer to the Research Question

Returning to our central research question: *How much biological detail does a neuron model need to reproduce a real neuron's electrophysiological behaviour, and which behaviours are preserved or lost when a detailed conductance-based model is reduced to a FitzHugh–Nagumo model?*

Our empirical and mathematical findings demonstrate that:

1. **For Subthreshold Passive Integration and Macroscopic Excitability Gating:** A dimensional two-variable FitzHugh–Nagumo model is **sufficient**. By calibrating eight parameters to a single sweep, FHN accurately reproduces passive membrane charging ($r^2 > 0.996$), matches input resistance, and reliably detects the boundary between silence and repetitive firing (92.5% sensitivity across 109 sweeps). In fact, FHN outperformed the published detailed models in excitability detection because the detailed models suffered from a $+60$ pA coarse rheobase overestimation in Populations A and C.
2. **For Spike-Train Timing, Rate Coding, and Waveform Kinetics:** The tested two-variable planar FHN formulation is **insufficient**. Compressing all voltage-dependent conductances and recovery processes into a single linear recovery variable (w) eliminates the dynamical degrees of freedom required to sustain multi-second spike-frequency adaptation, decouple onset latency from firing rate, capture rapid sodium upstroke kinetics, or reproduce hyperpolarization-activated inward sag.

12.2 The Hierarchy of Model Reduction

Our results clarify the boundaries across the hierarchy of computational neuron models:

$$\text{Multi-Compartment (NEURON)} \longrightarrow \text{Point Conductance (HH)} \longrightarrow \text{3D Adaptive (AdEx/Izhikevich)} \longrightarrow \text{2D Planar (FHN)}$$

The comparison between adaptive conductance models and the tested 2D planar cubic FHN formulation highlights the boundary where **multi-timescale adaptation and delayed onset decoupling are constrained**. In the tested autonomous two-variable cubic FHN formulation with linear recovery, sustaining adaptation across multi-second stimulus steps is structurally unavailable because the phase plane contains only a single slow recovery degree of freedom (w), causing limit cycle orbits to rapidly settle into stationary periodic firing (≤ 2 cycles). While certain non-polynomial 2D dynamical systems with subthreshold non-linearities or dynamic reset thresholds can generate transient slowing, capturing continuous multi-second adaptation and latency-frequency decoupling across experimental current-clamp steps provides strong scientific motivation for exploring model extensions that incorporate secondary slow processes (such as a second recovery variable w_2 or intracellular calcium accumulation $[Ca^{2+}]_i$ operating on a timescale $\tau \gg 100$ ms), rather than establishing a universal mathematical impossibility for all two-dimensional models.

12.3 Trade-offs Between Biophysical Realism and Computational Cost

Our evaluation highlights that neither model tier is universally superior across all electrophysiological criteria:

- **Detailed Multi-Compartment Models:** Excel at capturing action-potential waveform profiles (step RMSE 4–8 mV) and multi-timescale adaptation, but require high computational overhead, suffer from calibration sensitivity (e.g., Pop A $+13.6$ mV baseline offset), and can miss near-threshold excitability transitions.
- **Reduced FitzHugh–Nagumo Models:** Provide computational efficiency (simulating thousands of times faster via simple RK4 integration) and robust threshold detection via subcritical Hopf bifurcations, but sacrifice waveform fidelity and multi-timescale firing rate dynamics.

13. Limitations and Scientific Caveats

To ensure academic rigor, twelve explicit scientific caveats and boundaries must be documented:

1. **Representative Specimens ($n = 1$ per Layer):** Results characterize four individual biological specimens and their corresponding model fits. They do not constitute a statistically representative sample of entire cortical layers.
2. **Unconfirmed Projection Classes:** Layer-defined populations cannot be equated with functional projection classes (IT, ET, PT, CT) due to the absence of retrograde tracing metadata in the Allen dataset.
3. **Morphological Reconstruction Sourcing and Tissue Processing:** All four selected representative perisomatic models in Phase 3–5 utilize reconstructed 3D morphologies matching their respective recorded donor specimens (unlike exploratory candidate `382982932` in Phase 2, which required a substitute morphology due to missing Allen API files). Nevertheless, dendritic reconstructions from in vitro acute slices inherently experience tissue shrinkage, optical sectioning boundaries, and slicing truncations that introduce geometric approximations into multi-compartment passive cable calculations.
4. **Protocol-Dependent Biological Non-stationarity:** In Populations C (L5) and D (L6a), biological cells displayed protocol-dependent firing non-stationarity between COARSE and FINEST sweeps within the same recording session (e.g., Pop D firing 1 spike at +240 pA in FINEST vs. multiple spikes in COARSE).
5. **Layer 4 Late-Step Spike Cessation:** Specimen `484564503` ceased firing prematurely during strong 1000 ms depolarizing steps (+390 to +430 pA, COARSE protocol, 1020–2020 ms): at +410 pA (Sweep 51), the biological cell ceased firing at 1522.03 ms (13 recorded spikes; silent for the remaining 497.97 ms) vs. 39 detailed-model spikes (firing through 1992.51 ms); at +430 pA (Sweep 52), the biological cell ceased firing at 1498.34 ms (12 recorded spikes; silent for the remaining 521.66 ms) vs. 48 detailed-model spikes (firing through 2003.26 ms). The detailed model continued firing through the end of the pulse across all high currents. The biological mechanism responsible for this premature cessation remains unverified.
6. **Detailed-Model Rheobase Overestimation:** The $+60$ pA rheobase mismatch in Populations A and C remains an unresolved empirical observation; while calibration sweep amplitude selection is a plausible methodological contributor, it has not been causally isolated.
7. **FHN Adaptation Scope:** The distinction between transient initial ISI adjustments (≤ 2 cycles) and sustained multi-second adaptation must be preserved; FHN limit cycles are stationary to $< 0.01\%$.
8. **Biological Adaptation Census:** Continuous adaptation was observed across all 23 qualifying multi-cycle biological sweeps (≥ 3 spikes), with extreme adaptation ($\geq 5\times$) confined to 3 sweeps.
9. **Hyperpolarization Sag Provenance:** Biological sag at -110 pA was measured as modest (0.63–1.86 mV); detailed model sag was **not quantified** in evaluation metrics; FHN produced 0.00 mV under the tested condition.

10. **Provisional Action-Potential Kinetics:** Numerical upstroke velocities (397–466 V/s vs. 5.7–6.2 V/s) remain provisional supporting observations due to unstandardized filtering, sampling rates, and digital differentiation stencils.
11. **Subspace Boundedness of Population D Scans:** The latency-frequency trade-off in Population D is empirically demonstrated within the investigated 2D parameter subspace ($[40, 220]$ pF × $[8, 50]$ ms), not as an unproven universal theorem across all planar systems.
12. **Numerical Uncertainty of Bistability Intervals:** Fold currents (I_{fold}) and bistability widths are numerically estimated with an uncertainty of ± 0.01 pA based on step-refinement audits, and are not exact analytical proofs.

14. Conclusion

This project evaluated the electrophysiological consequences of model reduction by comparing intracellular cortical recordings, detailed multi-compartment biophysical models, and two-variable dimensional FitzHugh–Nagumo models across four layer-defined mouse visual cortex populations.

Our findings demonstrate that a two-variable dimensional FHN model provides an effective representation of **passive subthreshold charging ($r^2 > 0.996$) and macroscopic excitability gating (92.5% spike detection sensitivity)** via a verified subcritical Hopf bifurcation. However, reducing the biophysical state space to two planar variables results in the systematic loss of **sustained multi-second spike-frequency adaptation, fine action-potential waveform fidelity, hyperpolarization-activated inward sag, and latency-frequency decoupling**.

For large-scale neural network simulations prioritizing computational efficiency and macroscopic state transitions, the dimensional FHN model offers a calibrated alternative to abstract dimensionless formulations. However, for studies where spike-timing codes, sustained rate adaptation, or fine waveform kinetics are central, researchers should consider modelling extensions beyond the standard planar cubic formulation, such as incorporating secondary slow recovery variables, calcium-dependent adaptation processes, or fast-activating active conductances.

15. Future Work

Several promising avenues remain for extending this research:

1. **Three-Variable Adaptive Reductions:** Extending the dimensional FHN framework to a three-variable system (e.g., adding an activity-dependent adaptation current w_2 or slow calcium variable) to systematically evaluate whether a third dynamical variable restores sustained adaptation and latency decoupling.
2. **Standardized Kinetic Benchmarking:** Implementing a standardized digital filtering and differentiation pipeline (e.g., 5-point central difference stencils on 10 kHz filtered traces) to benchmark action-potential phase-plane dynamics ($\mathrm{d}V/\mathrm{d}t$ vs. V) across detailed and reduced models.
3. **Multi-Specimen Population Sampling:** Expanding the sample size from single representative specimens ($n = 1$) to multi-cell cohorts ($n \geq 20$) per cortical layer to quantify biological variance and statistical confidence intervals across cortical laminar sub-types.
4. **Higher-Dimensional Global Optimization:** Applying multi-objective genetic algorithms across all eight FHN parameters to investigate whether non-standard parameter regimes can achieve broader firing-rate ranges in Layer 6a without sacrificing latency.

16. References

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17. Reproducibility and Artifact Availability

All code, data pipelines, calibration algorithms, and evaluation scripts are maintained on the `ashish-work` branch of repository `Patel-Princy/INCM_project`.

17.1 Authoritative Repository Artifacts

- **Primary Synthesis Artifact:** `results/phase5/PHASE5_CROSS_POPULATION_SYNTHESIS.md`

- **Baseline Flash Handoff:** `results/phase5/PHASE5_BASELINE_AND_FLASH_HANDOFF.md`
- **Claim Audit Records:**
- `results/phase5/PHASE5B1_SCIENTIFIC_CLAIM_AUDIT.md`
- `results/phase5/PHASE5B2_EVIDENCE_DISCREPANCY_RESOLUTION.md`
- `results/phase5/PHASE5B3_FINAL_CONSISTENCY_CHECK.md`

17.2 Primary Data and Summary CSV Files

- Detailed Model Held-Out Metrics: `results/phase3b/heldout_metrics.csv`
- Detailed Model Rheobase Brackets: `results/phase3b/heldout_rheobase.csv`
- FHN Calibrated Parameters: `results/phase4b/fhn_fitted_parameters.csv`
- FHN Calibration Metrics: `results/phase4b/fhn_calibration_metrics.csv`
- FHN Held-Out Summary: `results/phase4c/population_summary.csv`
- FHN Held-Out Detailed Metrics: `results/phase4c/fhn_heldout_metrics.csv`
- Hopf Bifurcation Checks: `results/phase4c3/phase4c3_hopf_independent_checks.csv`
- Limit-Cycle Fold Verification: `results/phase4c3/phase4c3_fold_verification.csv`
- Open Bistability Verification: `results/phase4c3/phase4c3_bistability_verification.csv`
- Biological Adaptation Census: `results/phase4c3/phase4c3_biological_adaptation_scope.csv`
- Population D 625-Point Grid Scan: `results/phase4b1/fhn_population_d_pareto_audit.csv`
- Population D 961-Point Grid Scan: `results/phase4b2/fhn_population_d_grid_points.csv`

17.3 Automated Test Suite

The repository includes an extensive regression test suite containing **451 tests** verifying data integrity, numerical integration, bifurcation metrics, and scientific claims:

```
python -m pytest tests -q
# Status: 451 passed in 58.86s
```

Base commit baseline: `d6e1585` (Phase 4C.3.1).